

The Challenges and the Potential of MLLM-based Psychological Examination Tools

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Abstract—Although most psychological counseling and assessments are provided in person, they are subject to time and physical constraints. To explore the challenges and potential of AI for psychological examination, we developed *SimriSketch*, a MLLM-based screening tool for HTP Tests. To assess the feasibility of this approach, we conducted a user study with 67 participants, followed by interviews with six additional participants, including two expert art therapists. Findings show that *SimriSketch* is perceived as accessible and easy to use. However, participants noted that the test results were overly formal compared to the empathetic tone of human counselors, and some expressed limited trust in AI-generated outcomes. These findings suggest design implications for the development of MLLM-based psychological assessments that better align with the expectations of both examinees and professional therapists.

Index Terms—House-Tree-Person Test, Multimodal Large Language Model, Screening Tool, Art Psychological, Psychotherapy, Human-Computer Interaction

I. INTRODUCTION

Psychological counseling are traditionally conducted in face-to-face settings as it is important to ensure that patients' thoughts and emotions are well-delivered to the therapists for accurate assessment [1], [2]. However, in emergencies like the COVID-19 pandemic, conducting such face-to-face assessments may become difficult, potentially impacting the effectiveness of psychological therapy and evaluations. As a potential solution, several studies have explored remote tools for art psychological assessments [3], [4]. However, no immediate results are available as experts are needed for the assessment and write down the results.

To fill this gap, we designed *SimriSketch*, a web-based remote examination tool for the House-Tree-Person (HTP) test, which is a type of projective art psychological assessments, commonly used to examine personality traits [5], [6]. As shown in Fig. 1a, the system allows users to create house, tree, and person drawings on their own devices, including computers and smartphones, using a template-based interface in a web browser. Once submitted, the system automatically presents both brief and detailed interpretations of the rule-based results, as illustrated in Fig. 1b.

For the usability assessment of the tool, we conducted a user study with 67 participants, where they were asked to use the

system and provide their feedback. After using '*SimriSketch*', the participants completed Google Forms. Overall, they responded positively to the convenience of conducting projective tests such as the HTP online. However, several participants expressed concerns about the accuracy of the results.

As a potential solution to the poor accuracy of rule-based interpretations of *SimriSketch*, we conducted another user study with six participants including two expert art therapists to explore the potential and challenges of using Multimodal Large Language Models (MLLMs) for generating more precise assessments. Study 2 examined the evaluation of *SimriSketch*'s interpretations by both lay participants and experts. Lay participants used Zoom to request GPT interpretations of their own drawings, followed by brief interviews. Two experts assessed an HTP drawing produced outside the study, comparing GPT and professional interpretations. Findings showed that while facial expression analysis was weaker, other domains were largely comparable. Results suggested that interpretation quality may improve with continued training. Lay participants indicated a willingness to use AI if credibility is ensured, and experts also acknowledged its practical potential. As a result, MLLMs received positive feedback for the convenience of completing projective tests - such as the HTP - online and for enabling rapid access to desired information. However, they also noted limitations of poor recognition emotions from facial expressions and in the interpretation process.

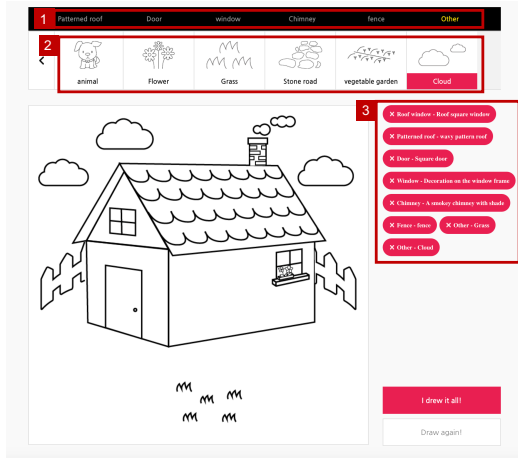
Drawing on user data and interviews with therapists, this paper characterizes the interaction qualities required for web-mediated early screening and organizes the design space of MLLM-mediated assessment with a focus on balancing interpretive trust and an empathetic tone. We suggest that integrating these interaction design implications into '*SimriSketch*', together with targeted model training, can further enhance the convenience and usability of psychological assessment tools.

II. RELATED WORK

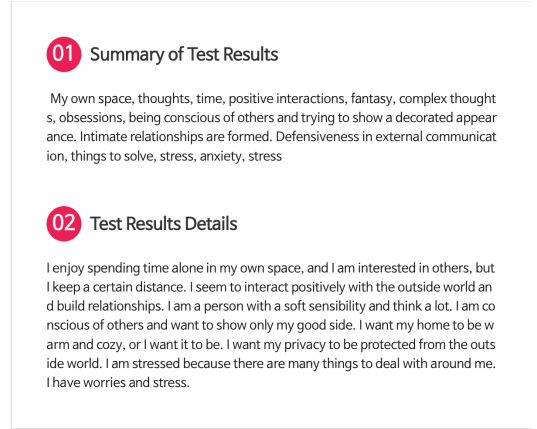
A. Web- and Mobile-based Art Assessment Applications

Recent work has explored web- and mobile-based platforms for psychological assessment, highlighting their potential to improve accessibility and scalability. Early digitized drawing-based assessments, such as the digital Clock Drawing Test (dCDT), showed that digital systems can capture not only final

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(a) Drawing screen



(b) Results

Fig. 1. Screenshot examples of *SimriSketch*, a multimodal large language model-based psychological examination tool with the HTP test: (a) the user interface for drawing a house, tree, and person where users can select (1) basic element categories such as *door* and *window*, (2) detailed elements (e.g., *a door with stairs*, *square door*) given selected basic element category, and (3) tags showing the selected detailed elements in the drawing; (b) the result page of the test.

outcomes but also process-level information, including stroke sequence and timing, which can support analysis of cognitive strategies and functioning [2], [7], [8]. Prior work also suggests that web-based assessments require careful consideration of usability, device compatibility, and latency, as input methods and platforms can affect performance measures [9]. These findings indicate that web-based art assessment tools should be designed with attention to accessibility, platform-independent validation, and robust interaction design.

B. Digital- and AI-based Evaluation Tools

Research on digital and AI-based tools for psychological assessment has expanded from feature-based drawing analysis to end-to-end image classification [7], [10]. In the HTP domain, existing studies have explored checklist-based methods, object detection, feature-based machine learning, and end-to-end classifiers [11]–[14], revealing a clear trade-off between interpretability and automation under Buck’s original HTP framework [6]. More recently, LLM- and MLLM-based approaches have been applied to drawings and other user-generated content for psychological assessment [15]–[17], with VS-LLM showing that stroke-level visual analysis combined with LLM-based semantic captioning can improve depression screening in PPAT [16]. While these studies demonstrate the promise of AI-assisted assessment, they also highlight the need for systems that balance interpretability, trust, and accessibility.

III. SIMRISKETCH

‘*SimriSketch*’ is a web-based screening tool for the HTP test. The name of our system is a combination of the Korean word ‘*Simri*’ (meaning ‘*psychology*’) and ‘*Sketch*’. This system is an online screening tool that allows users to take the HTP test at a time and place of their choosing. As shown in Fig. 1a, it improves accessibility by enabling users to construct drawings through selectable components rather than drawing everything by hand, while also providing immediate results.

A. Drawing Module

Users select one of three line types and then complete a picture in the click-based interface shown in Fig. 1a by choosing desired elements (e.g., a wavy-patterned roof, a fence). Selected elements are reflected in the drawing and displayed as removable chips. These visual elements were derived from previous studies and the NEO Personality Inventory [18], standardized and generalized, and limited to elements associated with meaningful interpretation results [19]. The completed drawing is then submitted for interpretation.

B. Interpretation Module

The drawings submitted are interpreted on the result screen shown in Fig. 1b. The interpretation is generated by a rule-based system comprising 118 rules in total (House = 35, Tree = 31, Person = 52), developed in consultation with HTP experts. For example, a smoky chimney with shade is interpreted in the summary as reflecting concerns due to family-related problems, and in the detailed result as suggesting signs of conflict and concern within the family. The system provides both summary and detailed results: the summary presents keywords for a quick overview of the user’s psychological state, whereas the detailed section offers sentence-level explanations of the user’s psychological characteristics.

IV. STUDY 1: USABILITY EVALUATION

To evaluate the usability of *SimriSketch*, we recruited 67 participants (18 males, 49 females) via social media, where they were asked to use *SimriSketch* remotely and provide feedback. Their average age was 32.39 ($SD = 7.10$) and none of them had background knowledge in art psychological.

A. Procedure

The study began by asking participants’ demographic information and their previous experiences with psychological tests (e.g., the types of tests they had taken, and the reasons

for taking the tests). Then, participants were instructed to visit our website where they can access *SimriSketch* using a device of their choice such as a desktop computer or a smartphone where they were asked to draw three images (i.e., House, Tree, Person) and submit their drawings to get both a brief and detailed versions of the test results. They were asked to read the results and provide their feedback including the evaluation of the system in terms of System Usability Score (SUS) [20]. All participants who successfully completed the survey were provided with the psychological assessment results from *SimriSketch* and approximately 4 USD worth coffee coupon.

B. Findings

Overall, participants were positive about the system in terms of perceived accuracy and satisfaction; the average score was 3.78 ($SD = 0.88$) and 3.97 ($SD = 0.80$), respectively where 5 is best. The SUS was also high as 81.57 which is considered as “EXCELLENT.” [21]. Here we summarize our quantitative results along with participants’ qualitative responses.

1) *Prior Experience with Psychology Tests*: To understand the general needs and interests in taking psychological tests, we asked prior experience. As a result, almost half of the participants had experience including MBTI [22] ($N = 38$), Sentence Completion Test (SCT [23], $N = 7$), HTP ($N = 5$), Bender-Gestalt Test (BGT [24], $N = 5$), Person in the Rain [25] ($N = 4$), and the Rorschach Inkblot Test [26] ($N = 3$). The most common reason for taking these tests was to have a better understanding of themselves ($N = 30$), while others said they had no particular reason ($N = 7$). We also asked how these tests were taken, they all reported online ($N = 35$), and 13 of them have taken the tests offline as well.

2) *Perceived Accuracy*: The perceived accuracy of test results was evaluated using a 5-point scale, along with the reasons for their ratings. Fifty participants felt that the results reflected many aspects such as specific details about their personality. On the other hand, 17 participants were not fully satisfied with the results for being inaccurate/ambiguous ($N = 10$), too general ($N = 3$). They also expressed concerns that the result could lead to similar or even identical interpretations as the drawings are created with only a few clicks of selection from the pre-defined template.

3) *Convenience*: We found that participants preferred an online method that allowed them to create drawings easily at a convenient time and place, rather than relying on face-to-face interactions to check their mental state. Participants ($N = 16$) expressed high satisfaction with the accessibility of the online test, noting that it could be used anytime and anywhere. Unlike traditional art-based assessments that are typically conducted offline, the online format enabled more convenient access. Participants noted that offline tests often require scheduling and travel, whereas the online test allowed them to participate at their preferred time and receive results immediately after submitting their drawings. This significantly reduced time and location constraints.

4) *Intuitiveness*: It was observed that users preferred an intuitive UI with a simple approach, making the system easy

for anyone to use ($N = 5$). They appreciated that, as shown in Fig. 1a, even if someone struggles with reading, they could easily choose what they wanted because each element for creating the drawing was accompanied by both an image and a name. Responses like “*With easy controls, anyone can easily participate in the psychological test.* (P42)” were observed.

5) *Expressiveness*: Participants expressed overall satisfaction with the ease of use of the system ($N = 46$). Many noted that being able to construct drawings by selecting options, rather than drawing them by hand, reduced the burden of participation ($N = 15$). At the same time, five participants reported limitations in expressiveness, particularly the restricted set of selectable elements and the difficulty of adjusting their size or position as desired. As P4 noted, although selecting predefined options reduced the burden of drawing, the limited range of available elements also made it difficult to express exactly what they had in mind.

V. STUDY 2: INTERVIEW

To investigate the potential and limitations of using MLLMs for interpreting HTP drawings, we conducted interviews with six participants: four non-experts (S1–S4) recruited from Study 1 and two counselors with HTP interpretation experience (E1 and E2). E1 reported conducting HTP interpretations 1–3 times per month, whereas E2 had prior experience with HTP but had not used it in practice during the past two years. Non-expert participants compared interpretations generated by *SimriSketch* in Study 1 and by an MLLM, while the expert group compared an MLLM-generated interpretation with one written by an independent expert.

TABLE I
INTERPRETATION OF FIG. 2 BY EXPERTS AND GPT-4O. THE BLUE UNDERLINED TEXT REPRESENTS INTERPRETATIONS THAT ARE SIMILAR OR OVERLAPPING (STABILITY, PROTECTION, POSITIVE ASPECTS, AND RESPONSIBILITY). THE RED TEXT REPRESENTS CONTRASTING PERSPECTIVES (NEGATIVE TENDENCIES VS. POSITIVE ORIENTATION, DEPENDENCE VS. RELATIONSHIP EMPHASIS).

Interpretation of a House Drawing	
Expert	“Your home appears to be <u>peaceful and stable</u> . You tend to overthink various aspects of family life, which can sometimes result in <u>obsessive tendencies</u> . At home, you seek relaxation and desire to protect your <u>privacy from external influences</u> . You have a <u>dependent inclination towards your family</u> and a desire to make a strong impression on others through active social interactions. Your family life is <u>stable</u> , and your interactions with others are handled with precision and control, maintaining a balanced approach. Some tasks and responsibilities require your attention, and you are managing these well. Overall, you take a <u>responsible approach</u> to tasks assigned to you within the household.”
GPT4o	“The <u>stable and functional structure</u> is well represented. The various elements around the house, such as the car or animals, and the detailed features of the house itself, may signal the things you consider important in your life. These detailed aspects can reflect a <u>positive attitude towards your current environment</u> and highlight the importance of relationships. However, it might also be worth considering whether this house is playing a role in <u>defending yourself from external influences</u> and how you perceive the structure of the house and its surroundings.”

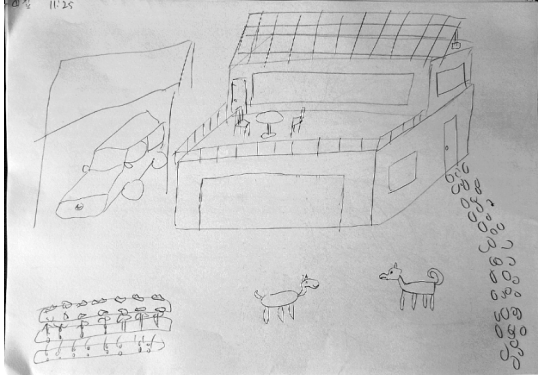


Fig. 2. Example house drawing used in Study 2. This pencil drawing was created based on the Task I interpretation results summarized in Table I and was provided to expert participants together with the corresponding Table I results for evaluation.

A. Apparatus

For the expert comparison, we used a house drawing and its interpretation provided by an independent expert, as shown in Fig. 2 and Table I, respectively. We also used OpenAI’s ChatGPT-4o [27] to generate an MLLM-based interpretation of the same drawing. To do so, the house drawing was uploaded to GPT-4o with the following prompt: “Hello! Can you interpret the HTP test for me? Please provide a psychological interpretation of the uploaded image based on the HTP test.”

B. Procedure

The procedure differed for the non-expert and expert groups. For the non-expert group, participants uploaded the three HTP drawings they had created in Study 1 and asked GPT-4o to generate interpretations using the same prompt. After receiving the interpretation, they were given 10 minutes to ask at least three open-ended questions about the result. They then compared the interpretation generated by *SimriSketch* in Study 1 with that generated by GPT-4o in terms of preference. Participants also rated the MLLM-based interpretation in terms of reliability, satisfaction, and accuracy using a 5-point Likert scale, and explained the reasons for their ratings. In addition, willingness to use an MLLM-based art assessment tool was assessed. The session was conducted online via Zoom, audio- and screen-recorded, lasted about 60 minutes, and participants received a coffee coupon worth approximately USD 7.

For the expert group, participants were given a set of four drawings (i.e., house, tree, man, and woman), accompanying question-and-answer sheets, and two interpretation results: one prepared by an independent expert and the other generated by GPT-4o. They were asked to evaluate the accuracy of the two interpretations without knowing which was which. This procedure was repeated for two additional drawing sets. Each expert session lasted approximately 120 minutes, and participants received approximately USD 40 in compensation.

C. Findings

Non-experts showed a positive response in terms of reliability, satisfaction, and willingness to use MLLM-based art

psychological tests. However, opinions were divided regarding the accuracy and the comparison between our system, *SimriSketch*, and the MLLM’s interpretation. Experts reacted more favorably to interpretations made by professionals and indicated that, while psychological experts do not have the intention to use MLLM-based art tests, non-experts may find them useful.

1) *Accuracy*: Among the non-experts, opinions on the accuracy of the interpretation were mixed. Two participants evaluated both the interpretation results and the responses to their follow-up questions positively, whereas two other non-experts expressed negative views, particularly regarding emotional interpretation. S4 felt that the interpretation did not align with their intended facial expression or current concerns, questioning whether “eyes alone can really indicate a smile.” The experts raised similar concerns, noting limitations in emotional analysis, particularly facial expressions, as well as in capturing line quality, erased and redrawn elements, and other contextual cues that may affect interpretation. These limitations made it difficult for them to regard GPT4o’s analysis as fully accurate, although the experts also noted that some interpretations were unexpectedly close to those of a human expert and suggested that its accuracy could improve further with continued learning.

2) *Reliability*: Although the interpretation results were generally received positively, trust in the analysis varied depending on the perceived reliability of the training data. Three non-experts expressed trust in the system, stating that they considered it reliable because it draws on a large amount of data and combines information from credible sources. By contrast, S4 questioned whether the data used to train the MLLM consisted only of factual information and therefore rated its reliability as average. Participants also noted that trust could vary depending on the tone of the response. S2 suggested that a softer and more familiar tone could help users feel more comfortable and build rapport more easily.

3) *Satisfaction*: Overall, responses were positive, reflecting not only satisfaction with the interpretation itself but also appreciation for the ability to immediately ask follow-up questions and clarify points of curiosity after reading the results. S1, S2, and S4 expressed satisfaction because GPT4o not only answered their specific questions but also provided additional explanations for other interpretations they were curious about, including reasons for why such results appeared. They particularly appreciated the detailed explanations across multiple interpretations. By contrast, S3 noted that although receiving answers was helpful, the explanations were sometimes overly detailed and made the interaction feel somewhat tedious. E2 raised a similar concern, noting that in actual counseling, counselors do not typically explain interpretations in terms of specific drawing elements, such as the roof or posture, as GPT4o does. However, both experts also noted that this approach could still be useful for people unfamiliar with such interpretations, including students and lay users.

4) *Preference*: Preferences between *SimriSketch* and GPT4o were divided: S1 and S4 preferred *SimriSketch*,

whereas S2 and S3 preferred GPT4o. S1, despite previously noting the high accuracy of GPT4o, preferred SimriSketch because its tone felt more comfortable and its interpretations more detailed and nuanced. By contrast, S3, who rated GPT4o relatively low in both accuracy and satisfaction because it was not an expert system, still preferred GPT4o because its responses were presented in a more organized format and made the basis of the interpretations easier to understand. This suggests that user preference may depend not only on perceived accuracy, but also on presentation style and tone.

5) *Intention to Use AI-Based Art Psychological Assessments*: While the use of MLLMs for interpreting art-based psychological tests may be limited in specialized professional settings, the results suggest their potential as simple screening tools. Participants reported a relatively high willingness to use AI-integrated art-based tests ($M = 4.00$, $SD = 0.71$). S2 noted that the accessibility of AI-based tests could support their use in contexts such as child psychological development education, where children may feel less burdened when participating. Participants also valued GPT4o for providing more detailed interpretations and immediate responses to follow-up questions, even though experts could better explain specific aspects of the results, such as posture in the person drawing or roots in the tree drawing (S1, S3, S4).

However, both user groups mentioned that GPT4o cannot be a replacement for assessment by human experts. Yet, E1 suggested that it could still be useful as a learning tool for trainees in psychotherapy or psychological examination, and both experts agreed that it could serve as a simple psychological reading tool for the general public. E1 also responded positively to combining the tone of a professional interpretation with GPT4o's tone, while participants suggested that allowing users to choose the response tone could improve comfort and usability.

VI. DISCUSSION

A. Reliability Challenges in Art Assessment Tools

Unlike the design proposed by Kim and Kang [3], our system adopts a rule-based, template-driven approach to improve ease of use while limiting expressive freedom. Although this design reduces the burden of drawing and allows participants to create images more easily, the constrained set of options may produce similar outputs across users. As a result, the drawings may not fully reflect individual characteristics, which could reduce the reliability and consistency of the resulting psychological interpretations.

These findings suggest the need for further research on how to improve the reliability of constrained art-based assessment tools. One promising direction is to complement structured drawing input with conversation-based question answering. For example, allowing users to respond to follow-up questions about their drawings through chatbots or conversational AI may provide richer contextual information and support more precise interpretations [28], [29]. In addition, protecting sensitive data, such as personal information and psychological states, is essential for fostering user trust [30], [31]. Building

on this, we propose a feedback system that protects users' personal attributes and information (e.g., age and gender) while still enabling psychological experts to review the submitted data. Further investigation is needed to understand how incorporating expert insight through such a system affects user trust. Ultimately, we suggest developing an MLLM-powered screening tool for art-based psychological assessment that combines conversational interaction, data protection, and expert validation to provide more accurate and reliable results.

B. The Impact of Delivery Style on Preference

Our research results confirmed [28], [32] that how information is provided to users can influence their preferences. Even when presenting the same results, we found that participants prefer different sentence structures and delivery styles, which affects how they perceive and accept the information, thus influencing their preferences. It is important to avoid repetitive phrases, overly complex sentence structures, and the use of strong language that can reduce readability. Furthermore, providing too detailed explanations during question responses can decrease readability and cause users to feel overwhelmed. This suggests that these factors should be important considerations when designing psychological assessment tools using MLLMs to conversation with users effectively.

C. The Impact of MLLMs on Interpretation Reliability

It was observed that participants' responses varied according to their prior experience with MLLMs. In particular, those who used MLLMs more frequently were more likely to question the accuracy of the responses and critically re-evaluate them. This pattern suggests that confidence in MLLMs-generated outputs may depend on the scope and quality of the data used for training, particularly in specialized domains. These findings highlight the importance of fine-tuning and further research on factors that shape user trust, such as reliable training data and expert feedback.

D. Potential of Online Use in Art Psychological Assessments

The findings indicate that participants preferred simple and familiar online psychological assessments and responded favorably to art-based approaches, as these formats supported a clearer understanding of their mental state. This underscores the importance of developing and using online screening tools to improve access to potentially complex art-based assessments. However, the limited options, fixed positions, and predefined sizes of template-based drawings may reduce interpretive accuracy. Although MLLMs show promise for improving psychological interpretation, current models still have limitations in analyzing specific visual cues, such as facial expressions. Digital tools can support faster identification of visual elements in drawings [11] and improve the reliability and validity of drawing-based assessments [33]. In addition, integrating MLLMs may streamline interpretation and provide users with more accurate and trustworthy results [16].

To address the remaining limitations, further research is needed, particularly on training MLLMs for emotion analysis,

including subtle facial expressions. Based on this, we propose follow-up work to evaluate an MLLM-based system that analyzes users' visual expressions and textual inputs for more accurate and efficient psychological interpretation.

VII. CONCLUSION

We propose SimriSketch, an online MLLM-based psychological examination tool with HTP test, and a user study with 67 participants showed that its perceived accuracy and satisfaction are high. A follow-up interviews revealed that the ability to ask follow-up questions and receive explanations of the drawing elements underlying the interpretations was valued, although it lacks detailed analysis, emotional understanding and domain-specific optimization. Still, the findings suggest that fine-tuned MLLMs could serve as simple psychological screening tools. Future work will expand the expressive freedom of the drawing interface, integrate fine-tuned MLLMs, and examine whether post-test follow-up questions improve interpretive accuracy.

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